

Skills Summary

Multiplying and Dividing Fractions Without Flip Mistakes

Use this reference while completing your worksheet or when studying.

Skill 1 — Cancel top with bottom, then multiply (cancel-then-multiply)

Rule: $\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$ (multiply straight across).

Before multiplying you may divide **one top number and one bottom number** by the same factor. That is legal because it is the same as reducing the answer afterwards. **Never** cancel two top numbers with each other, and never two bottom numbers — that changes the value.

Example: $\frac{5}{6} \times \frac{3}{10}$

Step 1. This is a multiplication, so I look for a top number and a bottom number that share a factor.

Step 2. 5 (top) and 10 (bottom) share 5; 3 (top) and 6 (bottom) share 3. That leaves $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

Try it: $\frac{4}{9} \times \frac{3}{8}$

check: $\frac{9}{1}$

Skill 2 — Flip only the divisor (flip-only-the-divisor)

Rule (keep, change, flip): $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$

Keep the first fraction unchanged, **change** $\div$ to $\times$, **flip** only the second fraction (the divisor). The reciprocal of $\frac{c}{d}$ is $\frac{d}{c}$, and $\frac{c}{d} \times \frac{d}{c} = 1$ — that is why flipping undoes the division. Cancel only *after* the flip.

Example: $\frac{2}{5} \div \frac{3}{4}$

Step 1. Dividing by a fraction means multiplying by its reciprocal, so the second fraction is the one I turn upside down.

Step 2. Keep $\frac{2}{5}$, change to $\times$, flip $\frac{3}{4}$: $\frac{2}{5} \times \frac{4}{3} = \frac{8}{15}$

Try it: $\frac{5}{8} \div \frac{1}{2}$

check: $\frac{5}{4}$

Skill 3 — Find and fix a wrong flip (find-and-fix-the-flip)

Two questions that catch every flip mistake:

1. Which fraction got turned upside down — the one *after* the division sign? 2. Is the answer on the right side of 1? Dividing by a fraction smaller than 1 makes the result **bigger** than the fraction you started with; dividing by a fraction bigger than 1 makes it smaller.

Example: Sam wrote $\frac{3}{5} \div \frac{2}{7} = \frac{5}{3} \times \frac{2}{7} = \frac{10}{21}$. Find and fix the error.

Step 1. I check which fraction was flipped: Sam turned over the first one, but only the divisor may be flipped.

Step 2. Correct: $\frac{3}{5} \times \frac{7}{2} = \frac{21}{10}$ — bigger than $\frac{3}{5}$, as it must be when dividing by a fraction less than 1.

Try it: Lee wrote $\frac{4}{7} \div \frac{2}{3} = \frac{7}{4} \times \frac{2}{3} = \frac{7}{6}$. What is the correct answer?

$\frac{4}{9}$:xəpə

(!) Watch out: Flipping belongs to division only — there is no flip in a multiplication problem. And cross-cancelling belongs to multiplication only: in a division, rewrite it as a multiplication *first*, then look for a top and a bottom to cancel.